PARENT ROUNDTABLE SERIES

Histamine, Sleep, and the Nervous System

What histamine does, why it can keep a child awake, and what families can do about it



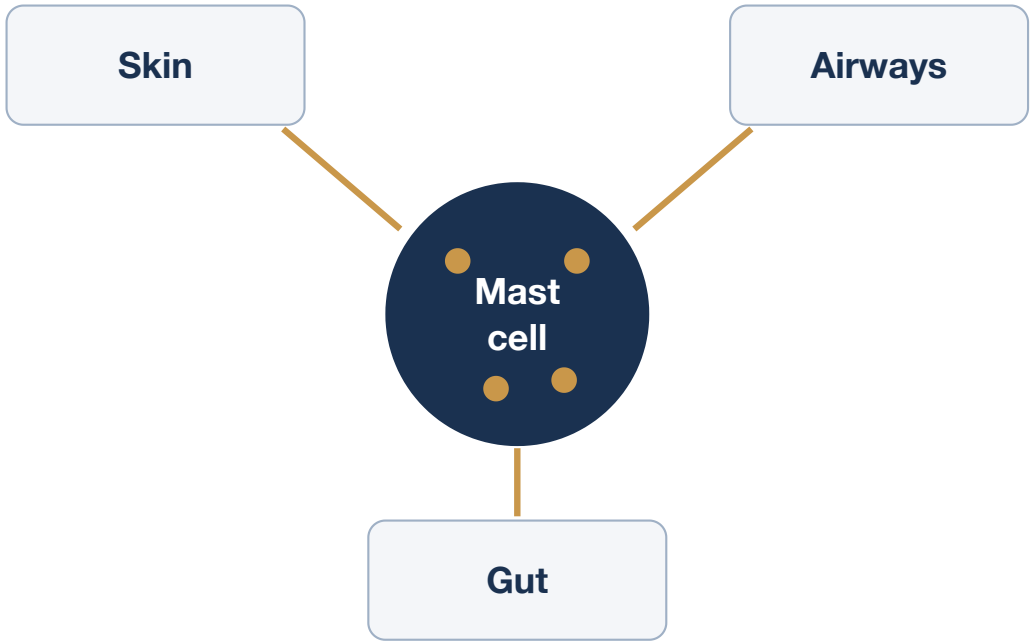
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WHAT HISTAMINE IS

Histamine is a messenger the body makes on purpose

- Stored in **mast cells** and **basophils**, ready to release in seconds
- Made by cells in the **stomach lining** to trigger stomach acid
- Used by **brain neurons** as a signal for alertness
- Mast cells sit where the body meets the outside world: skin, airways, and gut



Maintz L, Novak N. Am J Clin Nutr. 2007;85(5):1185-96. PMID 17490952

FOUR RECEPTORS, FOUR JOBS

The same molecule does very different jobs depending on its receptor

Receptor	Where it works	What it does
H1	Blood vessels, airways, skin, brain	Itch, swelling, and congestion in allergy; promotes wakefulness in the brain
H2	Stomach lining	Signals the stomach to make acid
H3	Brain and nerves	Fine-tunes the release of histamine and other neurotransmitters
H4	Immune cells	Directs immune cell movement in allergic inflammation

> This is why one family of medicines (H1 blockers) treats itch and sneezing, while another (H2 blockers) lowers stomach acid.

Thangam EB, et al. Front Immunol. 2018;9:1873. PMID 30150993

In the brain, histamine is a wake-up signal

- A small cluster of histamine neurons in the hypothalamus fires during waking and goes quiet during sleep
- Older antihistamines that cross into the brain block this signal, which is why they cause drowsiness
- When histamine runs high, a child can feel **wired and tired**: exhausted, but unable to settle



Haas H, Panula P. Nat Rev Neurosci. 2003;4(2):121-30. PMID 12563283

HOW THE BODY CLEARS HISTAMINE

Two enzymes clear histamine, and they need the right nutrients

DAO (diamine oxidase)

Breaks down histamine from food, working in the lining of the gut.

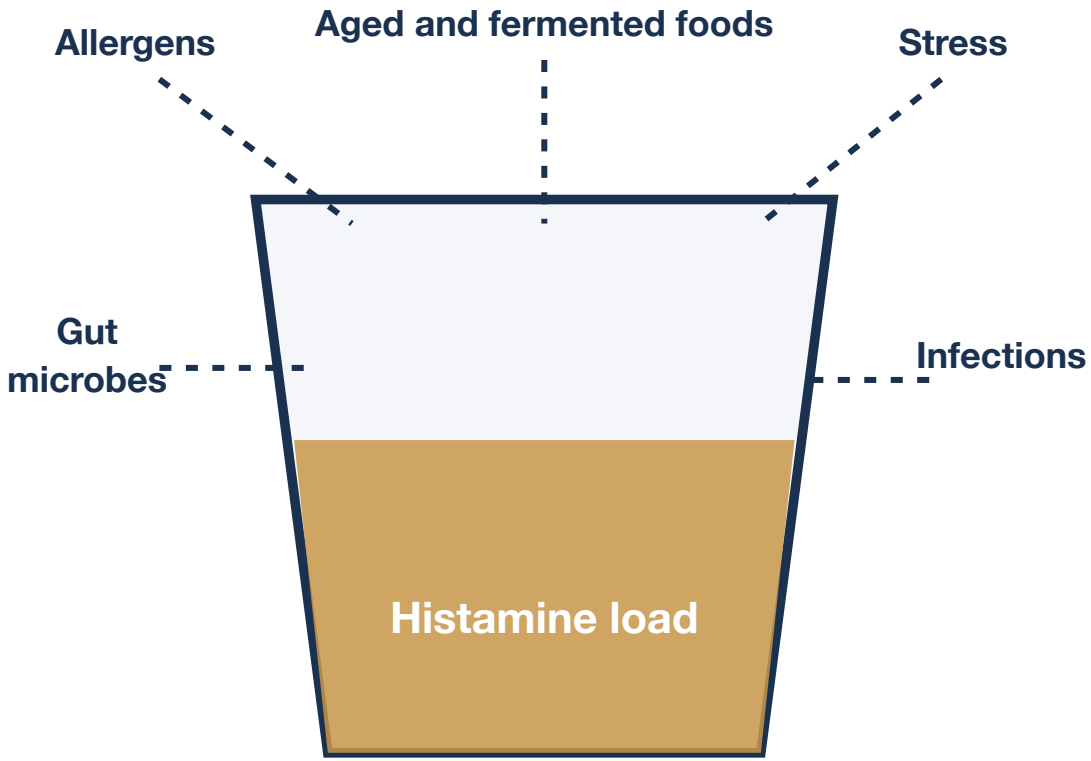
HNMT

Breaks down histamine inside cells throughout the body, including the brain.

Pharmacist lens: DAO is a copper-dependent enzyme, and vitamin B6 and vitamin C support histamine breakdown. Some medicines can also reduce DAO activity, so a medication review belongs in every histamine workup.

Maintz L, Novak N. Am J Clin Nutr. 2007;85(5):1185-96. PMID 17490952

Many sources fill the bucket; symptoms appear when it overflows



- Food, allergens, stress hormones, infections, and some gut microbes all add histamine
- DAO and HNMT drain the bucket
- A child may tolerate a food one week and react the next because the total load changed, not the food

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WHAT PARENTS CAN DO

Lower the load, support clearance, and protect sleep

Lower the load

- Track symptoms with meals, stress, and season
- Favor fresh foods over leftovers and aged foods during flares
- Reduce dust and mold at home

Support clearance

- Build meals around nutrient-dense whole foods
- Ask a clinician before adding supplements
- Bring a full medicine list to every visit

Protect sleep

- Keep a steady bedtime routine
- Keep the room cool, dark, and quiet
- Treat itch and congestion so they do not wake your child

> Talk with your pediatrician or pharmacist before using any antihistamine to help a child sleep.

SAFETY FIRST

Know when histamine symptoms need help right away

Call 911

- Swelling of the lips, tongue, or throat
- Trouble breathing or wheezing
- Hives with vomiting, dizziness, or fainting after a food, sting, or medicine

Use prescribed epinephrine first if your child has it.

Call your pediatrician

- Itch, rash, or congestion that disrupts sleep for more than a few nights
- Belly pain, diarrhea, or constipation that keeps returning
- Symptoms that change after starting a new food, supplement, or medicine

REFERENCES

Sources

1. Maintz L, Novak N. Am J Clin Nutr. 2007;85(5):1185-96. Histamine and histamine intolerance. PMID 17490952
2. Haas H, Panula P. Nat Rev Neurosci. 2003;4(2):121-30. The role of histamine and the tuberomamillary nucleus in the nervous system. PMID 12563283
3. Thangam EB, et al. Front Immunol. 2018;9:1873. The role of histamine and histamine receptors in mast cell-mediated allergy and inflammation. PMID 30150993